

2019 级高等数学 I(1) 考试卷(B)

参考解答及评分标准

一、填空题(本题共 6 小题, 每小题 5 分, 满分 30 分)

$$(1) \lim_{x \rightarrow 0} \left(\frac{e^x + 1}{2} \right)^{\frac{1}{x}} = \exp \left[\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} \right] = e^{\frac{1}{2}}.$$

$$(2) f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{(e^x - 1)(e^{2x} - 2) \cdots (e^{2020x} - 2020)}{x} = -2019!.$$

$$(3) \int_{-\pi}^{\pi} (\sin^3 x + \sqrt{\pi^2 - x^2}) dx = \int_{-\pi}^{\pi} \sqrt{\pi^2 - x^2} dx = \frac{1}{2} \pi^3.$$

$$(4) \text{通解为 } y = e^{-\int dx} \left[\int e^{-x} \cos x e^{\int dx} dx + C \right] = e^{-x} \left[\int \cos x dx + C \right] = e^{-x} (\sin x + C).$$

$$(5) I_2 = \int_0^{2\pi} e^{x^2} \sin x dx = I_1 + \int_{\pi}^{2\pi} e^{x^2} \sin x dx < I_1,$$

$$I_3 = \int_0^{3\pi} e^{x^2} \sin x dx = I_1 + \int_{\pi}^{2\pi} e^{x^2} \sin x dx + \int_{2\pi}^{3\pi} e^{x^2} \sin x dx = I_1 + \int_{\pi}^{2\pi} [e^{x^2} - e^{(\pi+x)^2}] \sin x dx > I_1,$$

故大小次序为 $I_2 < I_1 < I_3$.

$$(6) \text{齐次方程的特征方程为 } r^2 - 4r + 8 = 0, \text{ 根为 } r = 2 \pm 2i,$$

$$\text{通解形式为 } y = e^{2x} (C_1 \cos 2x + C_2 \sin 2x) + A e^{2x} + x e^{2x} (B \cos 2x + C \sin 2x),$$

其中 C_1, C_2 为任意常数, A, B, C 为待定常数.

二、解答题(本题共 5 小题, 每小题 7 分, 满分 35 分)

$$(1) \lim_{x \rightarrow 0} \frac{e^{x^2} - e^{2-2\cos x}}{x^4} = \lim_{x \rightarrow 0} \frac{e^{x^2-2+2\cos x} (e^{x^2-2+2\cos x} - 1)}{x^4} = \lim_{x \rightarrow 0} \frac{x^2 - 2 + 2\cos x}{x^4} \quad 3'$$

$$= \lim_{x \rightarrow 0} \frac{2x - 2\sin x}{4x^3} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{6x^2} = \frac{1}{12}. \quad 4'$$

$$(2) \text{两边求导得 } e^{-(x+y)^2} \left(1 + \frac{dy}{dx} \right) = \int_0^x \sin t^2 dt + x \sin x^2, \quad 4'$$

$$\text{令 } x=0 \text{ 得 } 1 + \frac{dy}{dx} \Big|_{x=0} = 0, \text{ 故 } \frac{dy}{dx} \Big|_{x=0} = -1. \quad 3'$$

$$(3) \int_1^3 f(x-2) dx = \int_{-1}^1 f(x) dx = \int_{-1}^0 (1+x^2) dx + \int_0^1 e^{-x} dx = \frac{7}{3} - \frac{1}{e}. \quad 2' + 2' + 3'$$

$$(4) \int_{\sqrt{2}}^{+\infty} \frac{1}{x\sqrt{x^2-1}} dx \stackrel{x=\sec t}{=} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{\sec t \tan t} \cdot \sec t \tan t dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} dt = \frac{\pi}{4}. \quad 5' + 2'$$

$$(5) \text{齐次方程的特征方程为 } r^2 - 2r + 1 = 0, \text{ 根为 } r_{1,2} = 1, \quad 2'$$

因 $\lambda=1$ 是特征方程的重根, 故特解形式为 $y^* = x^2(ax+b)e^x$,

$$\text{代入原方程得 } a = \frac{1}{6}, b = 0, \text{ 于是 } y^* = \frac{1}{6} x^3 e^x, \quad 2'$$

$$\text{所以原方程的通解为 } y = (C_1 + C_2 x) e^x + \frac{1}{6} x^3 e^x. \quad 3'$$

$$\text{三、设 } f(x) = (1+x) \ln^2(1+x) - x^2, \text{ 则} \quad 1'$$

$$f'(x) = \ln^2(1+x) + 2\ln(1+x) - 2x, \quad 4'$$

$$f''(x) = \frac{2\ln(1+x)}{1+x} + \frac{2}{1+x} - 2 = \frac{2}{1+x} [\ln(1+x) - x],$$

$$\text{从而 } f(x) = f(0) + f'(0)x + \frac{f''(\xi)}{2!} x^2 = \frac{1}{1+\xi} [\ln(1+\xi) - \xi] x^2 < 0 \quad (0 < \xi < x),$$

$$\text{即 } (1+x) \ln^2(1+x) < x^2. \quad 4'$$

$$\text{四、切线方程为 } Y - y = y'(X - x), \text{ 令 } X=0 \text{ 得 } Y_p = y - xy',$$

$$\text{法线方程为 } Y - y = -\frac{1}{y'}(X - x), \text{ 令 } Y=0 \text{ 得 } X_p = x + yy',$$

$$\text{由题意得微分方程 } y - xy' = x + yy', \quad 4'$$

$$\text{即 } \frac{dy}{dx} = \frac{\frac{y}{x} - 1}{\frac{y}{x} + 1}, \text{ 令 } \frac{y}{x} = u, \text{ 得 } u + x \frac{du}{dx} = \frac{u-1}{u+1}, \text{ 即 } \frac{u+1}{u^2+1} du = -\frac{dx}{x},$$

$$\text{两端积分得 } \frac{1}{2} \ln(1+u^2) + \arctan u = -\ln x + C,$$

$$\text{由 } x=1 \text{ 时 } u = \frac{y}{x} = 0 \text{ 得 } C=0, \quad 4'$$

$$\text{故所求曲线方程为 } \frac{1}{2} \ln \left(1 + \frac{y^2}{x^2} \right) + \arctan \frac{y}{x} = -\ln x \quad (x > 0),$$

$$\text{即 } \ln \sqrt{x^2 + y^2} + \arctan \frac{y}{x} = 0 \quad (x > 0). \quad 1'$$

$$\text{五、设切点为 } (x_0, \ln x_0), \text{ 则切线方程为 } y = \ln x_0 + \frac{1}{x_0} (x - x_0),$$

$$\text{由切线过点 } (0,1) \text{ 得 } x_0 = e^2, \text{ 故切点为 } (e^2, 2), \text{ 切线方程为 } y = 1 + \frac{x}{e^2} \text{ 或 } x = e^2(y-1), \quad 3'$$

$$A = \int_0^2 [e^y - e^2(y-1)] dy = e^2 - 1, \quad 3'$$

$$V = 2\pi \int_0^2 y [e^y - e^2(y-1)] dy = 2\pi \left(\frac{e^2}{3} + 1 \right). \quad 3'$$

$$\text{六、(1) 由 } \lim_{x \rightarrow +\infty} f(x) = 2 \text{ 得: } \exists X > 0, \text{ 使得当 } b > X \text{ 时 } f(b) > 1.5, \quad 2'$$

$$\text{由介值定理, } \exists a \in (0, b) \subset (0, +\infty), \text{ 使得 } f(a) = 1; \quad 2'$$

$$(2) \text{由拉格朗日中值定理, } \exists \xi \in (0, a), \text{ 使得 } f'(\xi) = \frac{f(a) - f(0)}{a - 0} = \frac{1}{a}. \quad 4'$$